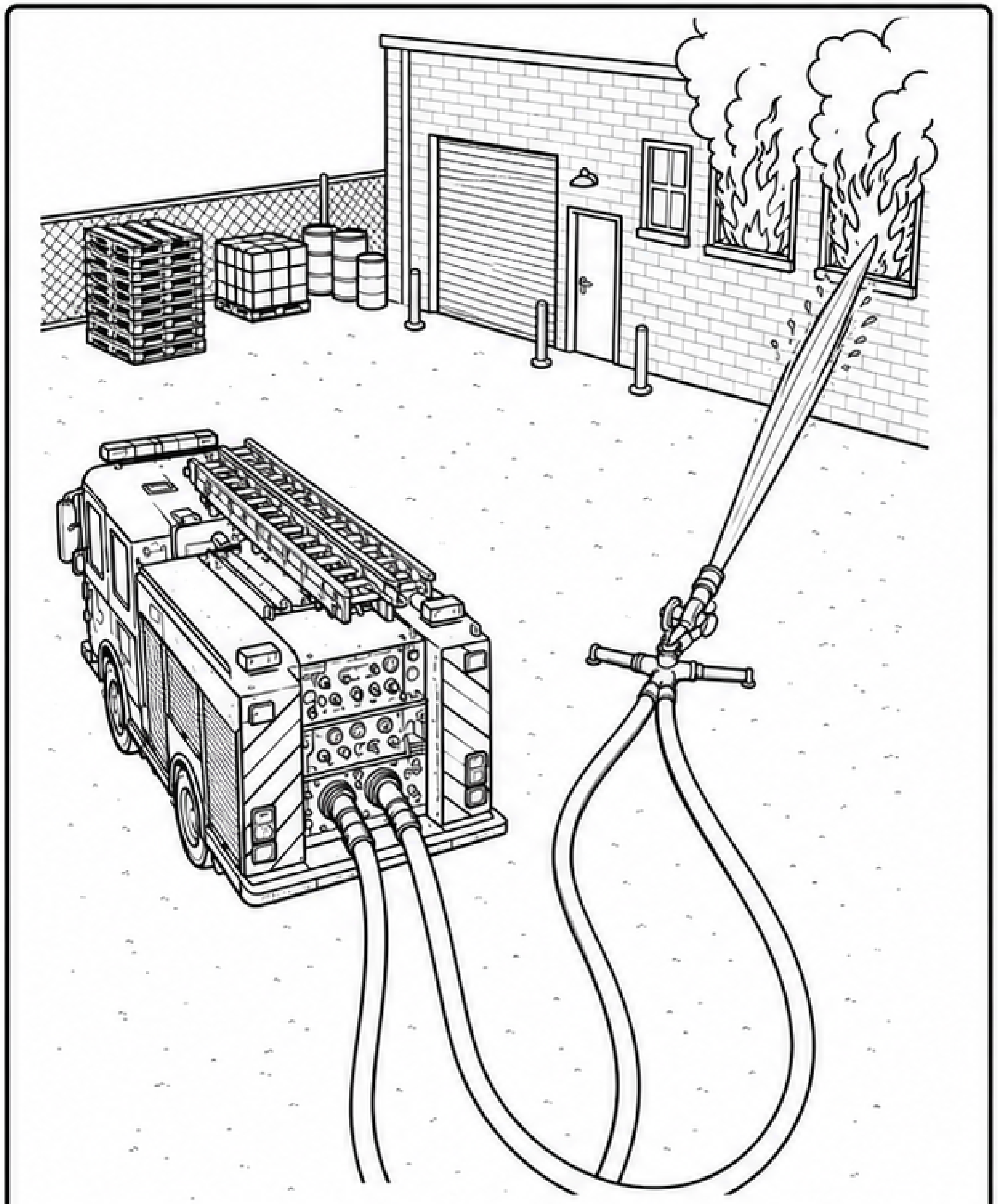
# FirePumpSim Printable Scenario

***Driver/Operator Practice Worksheet***

**Student Name:** ______________________ **Date:** ______________ **Class:** ______________ **Score:** __________

## Scenario: Industrial Building Fire - Portable Monitor Operation



### Given / Reference

- **Apparatus:** Engine 181
- **Operation:** Portable monitor supplying fire stream to 2nd-floor warehouse windows
- **Supply lines:** Two identical 200 ft of 2½-inch hose
- **Total monitor flow:** 500 GPM
- **Nozzle pressure (NP):** 80 PSI
- **Hose coefficient (C):** 2
- **Appliance loss:** 10 PSI
- **Elevation change:** 0 PSI

### Student Questions

**1.** The portable monitor is flowing 500 GPM through two identical supply lines. How many GPM is each 2½-inch supply line carrying?

______________________________

**2.** Using $FL = C \times Q^2 \times L$, what is the friction loss in each 200-foot 2½-inch supply line?

______________________________

**3.** What pump discharge pressure (PDP) is required to supply the portable monitor?

______________________________

**4.** If both supply lines were 250 feet long instead of 200 feet, what would the friction loss in each line be?

______________________________

**5.** If both supply lines were 250 feet long, what would the new pump discharge pressure (PDP) be?

______________________________

### Required Equations

- Q per line = Total GPM ÷ Number of equal lines
- $FL = C \times Q^2 \times L$
- Q = GPM / 100
- L = Hose Length / 100
- PDP = NP + FL + Appliance Loss ± Elevation

# FirePumpSim Answer Key

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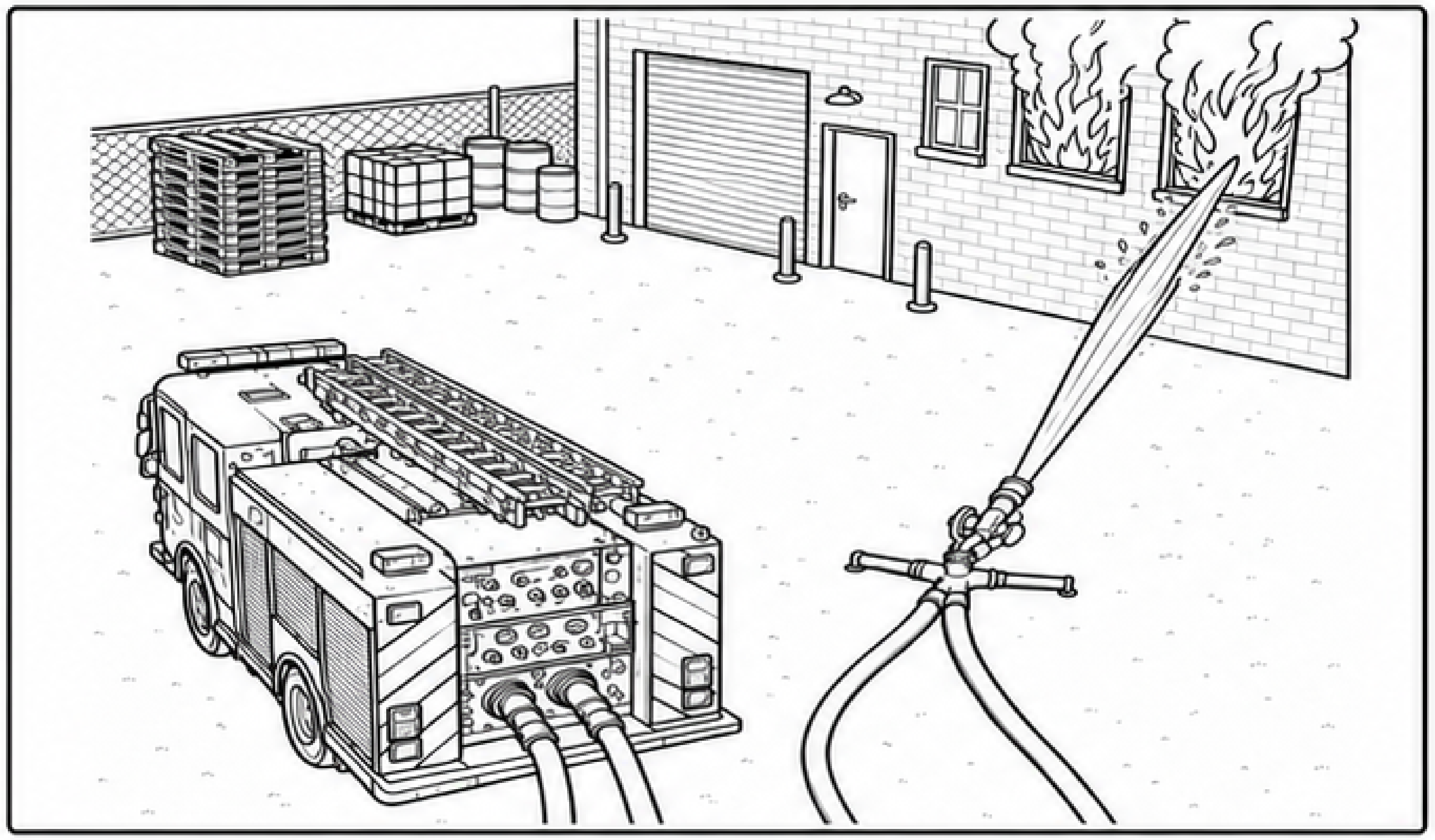

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## Instructor Answer Key

### 1. Flow in each 2½-inch supply line

**Formula:**
GPM per line = Total GPM ÷ Number of equal lines

GPM per line = 500 ÷ 2 = 250 GPM

Answer:
**250 GPM in each line**

### 2. Friction loss in each 200-foot 2½-inch supply line

**Formula:** $FL = C \times Q^2 \times L$

$Q = 250 / 100 = 2.5$

$L = 200 / 100 = 2$

$FL = 2 \times (2.5)^2 \times 2$

$FL = 2 \times 6.25 \times 2$

FL = 25 PSI

Answer:
**25 PSI**

### 3. Required pump discharge pressure (PDP)

**Formula:**
PDP = NP + FL + Appliance Loss + Elevation

PDP = 80 + 25 + 10 + 0

PDP = 115 PSI

Answer:
**115 PSI**

### 4. Friction loss if both supply lines were 250 feet long

**Formula:** $FL = C \times Q^2 \times L$

$Q = 250 / 100 = 2.5$

$L = 250 / 100 = 2.5$

$FL = 2 \times (2.5)^2 \times 2.5$

$FL = 2 \times 6.25 \times 2.5$

FL = 31.25 PSI

Rounded Answer:
**31 PSI**

### 5. New pump discharge pressure (PDP) with 250-foot lines

**Formula:**
PDP = NP + FL + Appliance Loss + Elevation

PDP = 80 + 31 + 10 + 0

PDP = 121 PSI

Answer:
**121 PSI**

**Accept reasonable rounding: ±1 PSI.**